

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Compare the incandescent lamp, fluorescent tube, CFL and LED with each other.
- Q.37 Draw and explain the electric circuit used in Refrigeration.
- Q.38 Draw and explain the block diagram of electrical locomotive and describe various equipments and accessories used.

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Q.1 Candela is the unit of
a) flux b) luminous intensity
c) illumination d) luminance

Q.2 In Arc heating the electrode are made of
a) Copper b) aluminum
c) graphite d) steel

Q.3 In a resistance furnace, the temperature is controlled by
a) variation of operating voltage
b) variation of resistance of heating circuit
c) switching on and off the supply periodically
d) all of these

Q.4 Electric arc welding use
a) DC supply
b) AC supply
c) Both DC and AC supply
d) None of these

Q.5 Electroplating is done for

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- a) replacement of worn-out material
 - b) protection of metals against corrosion
 - c) giving a shining appearance to articles
 - d) all of these
- Q.6 The capacity of a refrigerating machine is expressed as
- a) inside volume of the cabinet
 - b) lowest temperature attained
 - c) gross weight of machine in tones
 - d) rate of abstraction of heat from the space being cooled
- Q.7 D.C. motors are still preferred for use in
- a) electric excavators, steel mills and cranes
 - b) lathes and machine tools
 - c) flour mills and jaw crushers
 - d) paper industry
- Q.8 While plugging dc motos_____ connection are reversed.
- a) armature
 - b) field
 - c) both armature and field
 - d) auxiliary winding
- Q.9 Which of the following is used as the overhead wire of special cross section of traction system?
- a) conduction material
 - b) pantograph
 - c) catenary
 - d) smoothing reactor
- Q.10 Which of the following three phase ac system is used in electric traction system?

- a) 3 to 3.5 kV at 50 hz
- b) 3 to 3.5 kV at 25 hz
- c) 3 to 3.5 kV at 60 hz
- d) 3 to 3.5 kV at $16\frac{2}{3}$ hz

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Give two advantages of electric heating.
- Q.12 TIG stands for_____.
- Q.13 Bare electrodes are used for joining iron and mild steel. (True/False)
- Q.14 In galvanizing_____ metal is coated on steel.
- Q.15 Which type of motor is used in compressor of windows air conditioner.
- Q.16 During Rheostatic braking motor remains connected to supply mains. (True/False)
- Q.17 Define Scheduled speed.
- Q.18 Collection of current from trolley wire is done through_____.
- Q.19 Define Electric Drive.
- Q.20 Single to three phase system of electric traction is called_____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between direct and indirect resistance heating.
- Q.22 Write principle of Induction heating.
- Q.23 State laws of illumination.
- Q.24 Describe the construction of sodium vapour lamp.